

Assignment 2 Instructions

PSYC417

Assignment 2

This assignment is going to test your skills with functions, conditionals, and loops. You may get to the answer **using any of the techniques we specify in class, unless the question requests otherwise. If so, you must use that technique.** Do not use techniques or code we didn't learn in class (e.g., from a package we haven't used or a function we haven't covered). This will result in a deduction.

Description

The data for this assignment come from a greater correlational study on mental health. You will be working with just a few variables from the data set. Specifically:

- **sex**, biological sex, where only “Male” or “Female” were observed.
- **shs_1** through **shs_4**, which are responses to the four items of the Subjective Happiness Scale (Lyubomirsky & Lepper, 1999) where scores on items range from 1 through 7 and higher scores indicate greater happiness. (Note: This scale has one reverse-coded item, but this transformation has already been done on the data you see)
- **swl_1** through **swl_5**, which are responses to the five items of the Satisfaction with Life Scale (Diener et al., 1985) where scores on items range from 1-7 and higher scores indicate greater satisfaction with life.
- To load in the data, you may want to run the following, but **note** you will need to change the file path if the data and the assignment file are not in the same location:

```
a2_dat = read.csv(file = "a2_dat.csv")
```

Questions

Write each question/number into the Quarto document as text and place any code or text to answer the question on the lines below it. In short, it should look something like below:

Question 1 - Text from question 1. (MAKE THIS ONE OF THE HEADER STYLES SO IT'S EASIER TO READ.)

Code chunk, if applicable.

Text answering question, if applicable.

Question 2 - Text from question 2. (MAKE THIS ONE OF THE HEADER STYLES SO IT'S EASIER TO READ.)

.... and so on.

Question 0 (5pts total)

Create a Quarto document titled “Assignment 2” with you listed as the author (1pt). **Make sure the document type is .docx (1pt).** When uploading the assignment to ELMS, ensure you upload both the .qmd and .docx files (3pts). (No fixing this after the deadline if you forget.)

Question 1 (45pts total)

DO NOT USE ANY PRE-EXISTING COMMANDS LIKE VAR() ON QUESTION 1 UNLESS OTHERWISE STATED. You will need to hard-code most things.

Q1.a (25pts total):

Create a function called “sample_var” that computes the sample variance (s^2) of a vector. Remember, the formula for variance is:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

You must compute the variance by hand. You can use the **length()** function WHEN DEFINING THE LOOP, but not in the denominator of the variance calculation. (You have to use a manual approach, for example, see Assignment 1.) **YOU MUST USE A LOOP TO ANSWER THIS QUESTION.** You may use the **mean()** function to help solve this problem (i.e., you don't need to calculate the mean by hand). (10pts) Your sample_var function should have the following arguments:

- **data (5pts)**, which would accept a **numeric** or **integer** vector. Have it return an informative error message with any other data type. Generate your custom error message by entering an invalid object type or data type for value.
- **type (5pts)**, which either either accepts values of “manual” or “baseR”. Have “manual” be the default value for **type** and calculate the variance as described above. When set to “baseR”, have the function return the variance using the **var()** command. Have the function return an informative error message if this argument isn’t equal to “MANUAL” or “baseR”. Generate your custom error message by entering an invalid value for type.
- **digits (5pts)**, which rounds the variance to whatever the user enters for digits. **Have this round to 3 decimal places by default.** (Be sure this rounds for both “types” of variance calculation above. If **digits** is not set to a whole number, have it return an informative error message. (Hint: One way to check this is to use the **floor()** command, which removes any decimals after a number. So **floor(digits)** should equal **digits**, if **digits** is a number.) Generate your custom error message by entering an invalid digits value.

You may find the following psuedo code structure helpful:

```
# NAMEOFFUNCTION = function(ARGUMENTS){
#   if(DATA IS WRONG){
#     ERROR MESSAGE
#   } else {
#     if(TYPE IS WRONG){
#       ERROR MESSAGE
#     } else {
#       if(DIGITS IS WRONG){
#         ERROR MESSAGE
#       } else(
#         CALCULATE THE VARIANCE IN HERE
#       )
#     }
#   }
# }
```

Q1.b (10pts total)

Calculate the variance of the “sex” column using your function. Why did this generate an error?

Q1.c (10pts total)

Calculate the variance of the “shs_1” column using “baseR” as the type and assign it to an object called “bR_VAR”. Calculate the variance of the “shs_1” column using “manual” as the type and assign it to an object called “m_VAR”. Check whether bR_VAR and m_VAR are equal to each other using a conditional statement.

Question 2 (50pts total)

Q2.a (30pts total)

Create a function called “tot_scores” which calculates the total scores for a psychological survey. With this function, we want a DATA FRAME consisting of the scores for a single scale. The function should have two arguments:

- **data (15pts)**, which will hold the data frame. Check to see that **data** is a data frame, and return an informative error message if not. Additionally, make sure each column of **data** is numeric or integer and return an informative error message if not.
- **type (15pts)**, which can either be set to “mean” or “sum”. Have it calculate an average score or a sum score depending on what the user sets the value to, and have it return an informative error message if it’s set to anything else. (Hint: You may need the rowSums() function for calculating sum scores or the colMeans() function for calculating average scores. Look into whether you need to use the na.rm = TRUE argument in the colMeans() and rowSums() functions!)

Some practice code to get you set up is below:

```
# FUNCTION_NAME <- function(ARGUMENTS){ # open function
#   if(DATA IS NOT DATA FRAME){
#     ERROR MESSAGE
#   } else {
#     if(COLUMNS ARENT ALL NUMERIC OR INTEGER){
#       ERROR MESSAGE
#     } else{
#       if(TYPE IS NOT MEAN OR SUM){
#         ERROR MESSAGE
#       } else{
#         CALCULATE THE SUM OR AVERAGE SCORES DEPENDING ON WHAT THEY SET THE VALUE TO
#       }
#     }
#   }
# }
```

Q2.b (10pts total)

Add a column to the a2_dat data frame called “SHS_MEAN” and assign it the value of a mean score for the 4 shs items using your function. This should be a direct process (i.e., create the column and set it equal to the function output all in one line, no semi-colons). Tip: Remember to index the data frame for the correct items.

Q2.c (10pts total)

Add a column to the a2_dat data frame called “SWL_SUM” and assign it the value of a sum score for the 5 swl items using your function. This should be a direct process (i.e., create the column and set it equal to the function output all in one line, no semi-colons). Tip: Remember to index the data frame for the correct items.

The End

Render the document and upload the .qmd and .docx files to ELMS.

Thank you! :)